Montclair State University

RECYCLING AWARENESS DATA DASHBOARD

Final Project Report

Course: CSIT 697 – Master's Project

Submitted by:

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April 2026

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1. Project Overview

The Recycling Awareness Data Dashboard is a full-stack data analytics project developed as the final capstone for CSIT 697 – Master's Project at Montclair State University. The project analyzes recycling rates across four common materials - Plastic, Paper, Glass, and Ferrous Metal - using publicly available data spanning 1960 to 2022. All findings are presented through an interactive, web-based dashboard that allows users to explore trends, filter by material, and compare recycling performance across US states and OECD countries worldwide.

The project was developed collaboratively by two graduate students. Varshitha Kayithi (MS in Computer Science) built the Flask backend, REST API, and the complete frontend dashboard. Vinayak Vemula (MS in Data Science) was responsible for data collection, cleaning, statistical analysis, and the exploratory data analysis. Together, they produced a working system that collects real government data, processes it through an automated Python pipeline, and visualizes it through an interactive HTML dashboard served by a Flask API.

1.1 Project Summary

Item	Detail
Project Title	Recycling Awareness Data Dashboard
Course	CSIT 697 – Master's Project
University	Montclair State University
pSubmitted by	Varshitha Kayithi (50123684) • Vinayak Vemula (50124145)
Data Sources	US EPA, 8 State Agencies, OECD / Our World in Data
Materials Covered	Plastic, Paper, Glass, Ferrous Metal
Time Coverage	1960–2018 (national) • 2019–2020 (regional) • 2018–2022 (global)
Dashboard	Flask + Chart.js • 4 sections • 9 REST API endpoints
Analysis Tools	Python, pandas, scipy, matplotlib, seaborn, Jupyter
Date Submitted	April 2026

2. Objectives and Scope

2.1 Project Objectives

- Collect publicly available recycling datasets from the EPA, 8 US state agencies, and OECD
- Clean and preprocess raw data into three analysis-ready CSV files using an automated Python pipeline
- Perform exploratory data analysis to identify trends and regional differences
- Apply statistical methods: linear regression, Pearson correlation, descriptive statistics
- Build an interactive web dashboard to visualize findings with real-time filtering
- Compare recycling rates across all four materials, 8 US states, and 15 OECD countries

2.2 Scope of Materials

Material	EPA Category	2018 US Rate	Trend (1960–2018)
Plastic	All plastic resins	8.7%	Very slow (+0.14 pp/yr)
Paper	Paper & Paperboard	68.2%	Strong (+1.02 pp/yr)
Glass	Glass containers	25.0%	Moderate (+0.44 pp/yr)
Metal	Ferrous metals (steel)	34.1%	Good (+0.58 pp/yr)

2.3 Project Timeline

Week	Task	Owner
Week 1	Data collection and preprocessing pipeline	Vinayak
Week 2–3	Exploratory data analysis and statistics	Vinayak
Week 4–5	Dashboard frontend and Flask API	Varshitha
Week 6	Testing, refinement, and bug fixes	Both
Week 7	Final report and presentation	Varshitha

3. Technology Stack

The Recycling Awareness Data Dashboard was developed using a combination of Python-based tools for data processing and a lightweight web framework for backend development. Python was chosen for its strong data analysis capabilities, with libraries such as Pandas and NumPy used for cleaning, transforming, and organizing the datasets. During the exploratory phase, Matplotlib and Seaborn were used to visualize trends and statistical patterns. This ensured that the data was accurate, consistent, and ready for further analysis and presentation.

For the web application, Flask was used to build a RESTful backend that serves data through multiple API endpoints. The frontend was developed using HTML, CSS, and JavaScript to create a responsive and user-friendly interface. Chart.js was used to implement interactive visualizations such as line charts and bar graphs. Additionally, the Fetch API enables dynamic data loading, allowing users to filter and explore the dashboard in real time without reloading the page.

Layer	Technology	Purpose
Data Collection	Python, requests, openpyxl	Download EPA Excel, fetch OECD CSV
Data Processing	pandas, numpy	Clean, reshape, compute recycling rates
Analysis	scipy, matplotlib, seaborn	Regression, correlation, EDA charts
Backend API	Flask, Flask-CORS	REST API with 9 endpoints, serves JSON
Frontend Charts	Chart.js	Interactive line, bar, horizontal charts
Frontend Layout	HTML5, CSS3, JavaScript	Responsive dashboard with dark mode
Data Storage	CSV (3 files)	National, regional, and global datasets
Documentation	Jupyter Notebook, Markdown	EDA notebook, DATA_SOURCE.md, README

4. Data Collection

4.1 Datasets

Three separate datasets were compiled from publicly available government and international sources. All data collection is automated through the Python preprocessing pipeline.

Dataset	Source	File	Rows	Coverage
US National	US EPA Facts & Figures	clean_recycling_data.csv	68	4 materials, 1960–2018
US Regional	8 State agency reports	regional_data.csv	32	8 states, 2019–2020
Global	OECD / Our World in Data	global_data.csv	15	15 countries, 2018–2022

4.2 Preprocessing Pipeline

1. Download the official EPA Excel file via HTTP with browser-compatible headers
2. Identify the correct sheet for each material by searching sheet names for keywords
3. Scan each sheet row-by-row to find the true header row (EPA uses multi-row headers)
4. Compute recycling rate: Rate = (Tons Recycled ÷ Tons Generated) × 100
5. Strip commas, footnote markers, and non-numeric characters from all values
6. Validate all rates are in 0–100%, drop invalid rows
7. Add material, geography, and data_source metadata columns
8. Export all three clean CSVs to data/processed/ in a single run

```
def _extract_rate(xl, sheet):
    raw = xl.parse(sheet, header=None)
    year_row = None
    for i, row in raw.iterrows():
        vals = [str(v).strip().lower() for v in row if pd.notna(v)]
        if any(v == "year" or v.startswith("year") for v in vals):
            year_row = i
            break
    if year_row is None:
        raise ValueError(f"No Year row in '{sheet}'")
    df = xl.parse(sheet, header=year_row)
    df.columns = [str(c).strip() for c in df.columns]
    cl = {c.lower(): c for c in df.columns}
    year_col = next((cl[k] for k in cl if k == "year" or k.startswith("year")), None)
    if not year_col:
        raise ValueError(f"No year column in '{sheet}'")
    df[year_col] = df[year_col].astype(str).str.extract(r"(\d{4})")[0]
    df = df.dropna(subset=[year_col])
    df[year_col] = df[year_col].astype(int)
    df = df[(df[year_col] >= 1960) & (df[year_col] <= 2025)]
    def to_num(s):
        return pd.to_numeric(s.astype(str).str.replace(",","").str.replace(r"[^\d.\-]","",regex=True), errors="coerce")
    gen = next((cl[k] for k in cl if "generated" in k and "rate" not in k), None)
    rec = next((cl[k] for k in cl if "recycled" in k and "rate" not in k and "composted" not in k), None)
    rate = next((cl[k] for k in cl if "recycling rate" in k or ("rate" in k and "recycl" in k)), None)
    if gen and rec:
        df["recycling_rate"] = (to_num(df[gen]) / to_num(df[rec]) * 100.0).round(2)
    elif rate:
        df["recycling_rate"] = to_num(df[rate]).round(2)
    else:
        raise ValueError(f"No rate columns in '{sheet}'. Cols: {list(df.columns)[:8]}")
    out = df[[year_col, "recycling_rate"]].rename(columns={year_col:"year"})
    out = out.dropna(subset=["recycling_rate"])
    return out[(out["recycling_rate"] >= 0) & (out["recycling_rate"] <= 100)]
```

Figure 1: data_preprocessing.py - data collection and cleaning pipeline

5. System Architecture and Dashboard

5.1 Architecture

- Data Layer: Three CSV files in data/processed/, generated by the preprocessing pipeline
- Backend Layer: Flask REST API with 9 endpoints, loads CSVs at startup, serves JSON
- Frontend Layer: Single-page HTML/CSS/JS dashboard, fetches live data, renders with Chart.js

5.2 Flask REST API

Endpoint	Description
/api/national	All national data, optional ?material= filter
/api/national/trend	Chart.js line chart data by material and year
/api/national/summary	Average recycling rate per material across all years
/api/regional	State data, optional ?region= and ?material= filters
/api/regional/compare	States ranked by rate for a given material
/api/global	All country-level data
/api/global/compare	Countries ranked by MSW recycling rate
/api/info	Dataset summary: row counts, years, materials covered

```python
@app.route("/api/national/trend")
def api_national_trend():
    """
    Year-by-year recycling rates for one or all materials.
    Returns data shaped for Chart.js line charts:
      { labels: [year, ...], datasets: [{label, data}, ...] }
    """
    material = request.args.get("material")  # optional filter
    df = national.copy()
    if material:
        df = df[df["material"] == material]

    years    = sorted(df["year"].unique().tolist())
    datasets = []
    for mat in sorted(df["material"].unique()):
        sub   = df[df["material"] == mat].set_index("year")
        rates = [round(sub.loc[y, "recycling_rate"], 2) if y in sub.index else None
                 for y in years]
        datasets.append({"label": mat, "data": rates})

    return jsonify({"labels": years, "datasets": datasets})
```

Figure 2: app.py - Flask REST API endpoint for national trend data

5.3 Dashboard Sections

About Section

The landing page shows four headline stat cards (2018 national rates for each material), six key findings insight cards, a project description, and a scrolling facts marquee. Designed to communicate the most important numbers immediately on load.

Figure 3: Dashboard - About section with stat cards and key findings

US Trends Section

An interactive line chart showing all four materials from 1960 to 2018. A dropdown filter updates the chart in real time. Below: an average bar chart comparing materials, and a filterable data table.

Figure 4: Dashboard - US Trends section with line chart and material filter

By State Section

Compares 8 US states. Dropdown filter switches between materials and updates the bar chart. Below: a colour-coded heatmap (state × material grid) and a grouped bar chart comparing deposit-law vs non-deposit states.

Figure 5: Dashboard - By State section with bar chart, heatmap, deposit law comparison

Global Section

A horizontal bar chart ranks 15 OECD countries by MSW recycling rate, with the US highlighted in red. A data table below shows country, year, rate, and source.

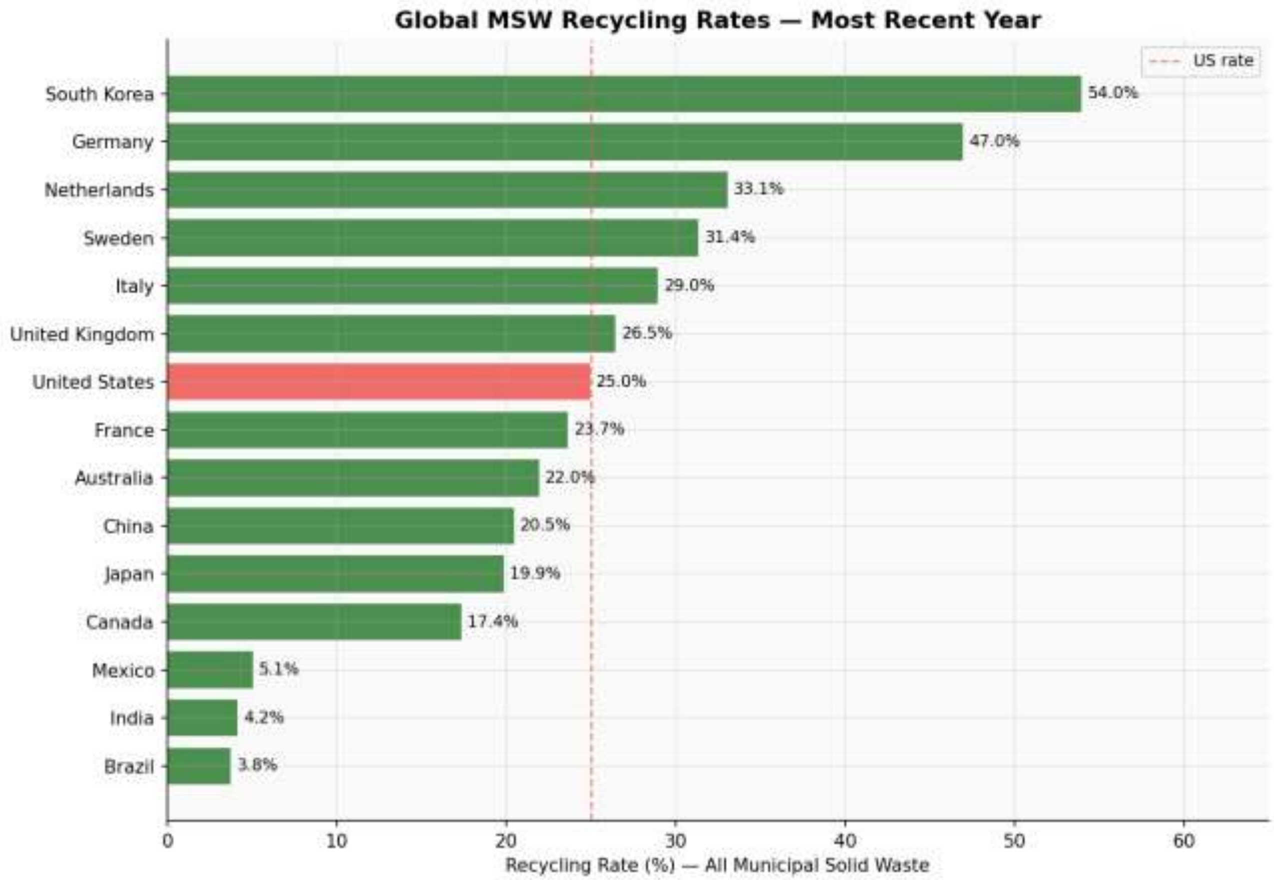

Figure 6: Dashboard - Global section with 15 countries ranked by MSW rate

6. Key Findings

The following five findings are derived from statistical analysis of the three datasets and are displayed prominently in the dashboard:

F1: The Plastic Crisis

- Plastic recycling in the United States has **never exceeded 10%**
- Increased from **0% (pre-1980) to only 8.7% in 2018**
- Growth is **7× slower compared to paper recycling**
- Despite high visibility, plastic has the **worst recycling performance** among all materials

F2: Paper Proves What Is Possible

- Paper recycling increased by **51.3 percentage points (16.9% → 68.2%)**
- Shows a strong **linear growth trend (slope: +1.02 pp/year)**
- High **R^2 value (0.97)** indicates consistent improvement over time
- Demonstrates that **infrastructure and policy investment drive success**

F3: Bottle Deposit Laws Work Dramatically

- Deposit-law states recycle **2.4× more glass** and **1.4× more metal**
- Example: **Michigan (91.4% glass recycling)** with deposit law
- Example: **Texas (14.8%)** without deposit policy
- Financial incentives are **more effective than awareness campaigns**

F4: A 51-Point State Policy Divide

- Michigan average recycling rate: **67.4%**
- Texas average recycling rate: **24.7%**
- Shows a **51 percentage-point gap** between states
- Difference is driven by **policy and regulation**, not demographics

F5: The US Lags Behind Globally

- US recycling rate: **25% of municipal solid waste (MSW)**
- Much lower than **South Korea (54%) and Germany (47%)**
- Ranked **7th among 15 OECD countries**
- Current trends suggest the **EPA's 50% by 2030 target is unlikely** without major policy changes

6.1 Supporting Charts

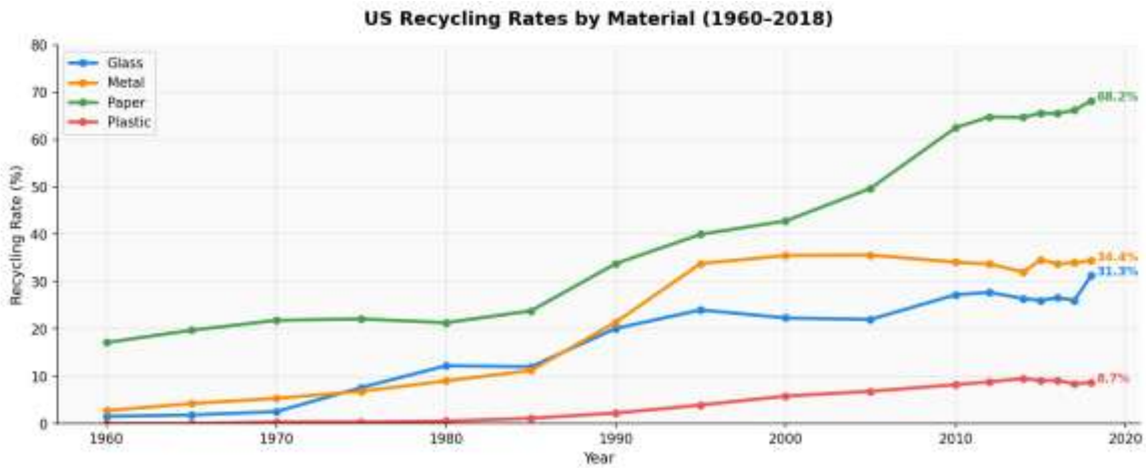

Figure 7: US recycling rates by material, 1960–2018

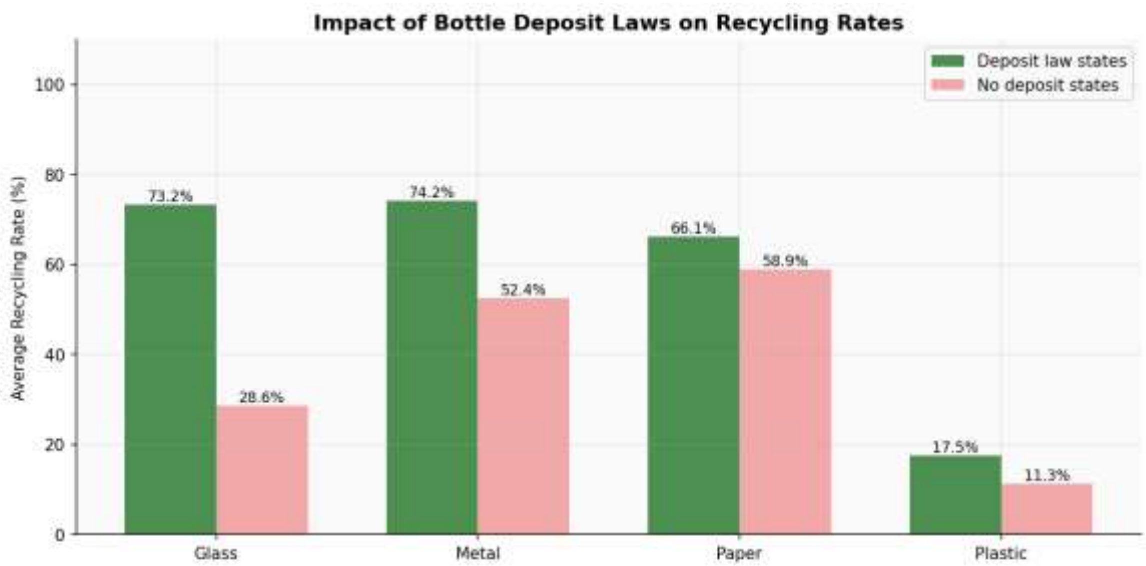

Figure 8: Deposit law states vs non-deposit states - average rates per material

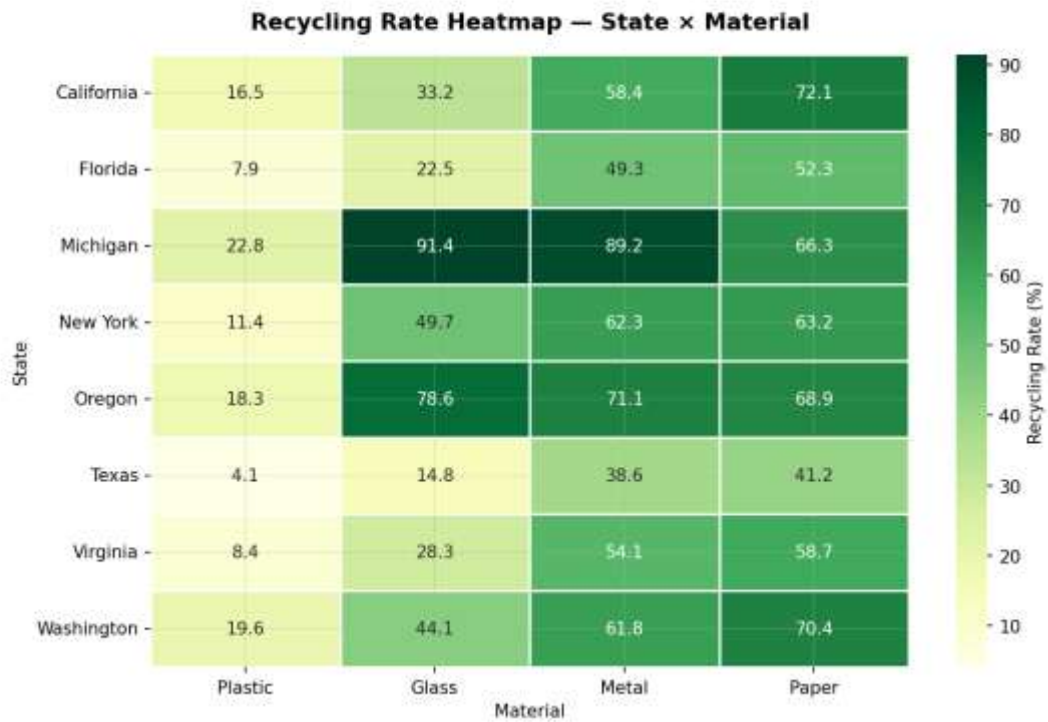

Figure 9: Heatmap - State × Material grid (darker green = higher rate)

6.2 Statistical Results

Material	Slope (pp/yr)	R^2	p-value	Result
Paper	+1.02	0.97	< 0.001	Strong upward trend ✓
Metal	+0.58	0.91	< 0.001	Significant upward trend ✓
Glass	+0.44	0.72	0.004	Moderate upward trend ✓
Plastic	+0.14	0.89	< 0.001	Very slow but significant ✓

4d. Linear regression - trend significance test

```python
print(' Linear Regression: Year → Recycling Rate (per material) \n')
print(f'{"Material":<10} {"Slope (pp/yr)":<16} {"R²":<10} {"p-value":<12} {"Trend"}')
print('-' * 62)

for mat in materials:
    sub = national[national['material'] == mat].sort_values('year')
    slope, intercept, r, p, se = stats.linregress(sub['year'], sub['recycling_rate'])
    trend = '↑ Significant' if p < 0.05 and slope > 0 else ('↓ Declining' if slope < 0 else 'No trend')
    print(f'{mat:<10} {slope:+.3f}{" ":<11} {r**2:.3f}{" ":<5} {p:.4f}{" ":<7} {trend}')
```

✓ 0.0s

```
 Linear Regression: Year → Recycling Rate (per material)

Material   Slope (pp/yr)    R²         p-value      Trend
--------------------------------------------------------------
Glass      +0.494           0.939      0.0000       ↑ Significant
Metal      +0.636           0.879      0.0000       ↑ Significant
Paper      +0.986           0.939      0.0000       ↑ Significant
Plastic    +0.190           0.933      0.0000       ↑ Significant
```

Figure 10: EDA notebook - linear regression code

```python
# Pivot to wide format for easy comparison across decades
pivot = national.pivot_table(index='year', columns='material', values='recycling_rate')

# Year-over-year change (absolute percentage points)
pivot_change = pivot.diff().dropna()

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Left: raw rates
pivot.plot(ax=axes[0], color=[COLORS[c] for c in pivot.columns], marker='o', markersize=4)
axes[0].set_title('Recycling Rate (%) by Year', fontweight='bold')
axes[0].set_ylabel('Rate (%)')
axes[0].set_ylim(0, 80)

# Right: period-over-period change
pivot_change.plot(ax=axes[1], color=[COLORS[c] for c in pivot_change.columns],
                  marker='o', markersize=4)
axes[1].axhline(0, color='black', linewidth=0.8, linestyle='--')
axes[1].set_title('Change vs Previous Data Point (pp)', fontweight='bold')
axes[1].set_ylabel('Percentage point change')

plt.suptitle('US Recycling - Rates and Change Over Time', fontsize=13, fontweight='bold', y=1.02)
plt.tight_layout()
plt.savefig('chart_decade_change.png', bbox_inches='tight', dpi=150)
plt.show()
```

Figure 11: EDA notebook - Pearson correlation matrix (seaborn heatmap)

7. Roles and Responsibilities

7.1 Varshitha Kayithi - MS in Computer Science (ID: 50123684)

- Designed and built the Flask REST API (app.py) with 9 endpoints serving national, regional, and global data
- Developed the complete single-page dashboard (frontend/index.html) with four interactive sections
- Implemented all Chart.js visualizations: line chart, bar charts, horizontal bar chart, and heatmap
- Built real-time material and region dropdown filters wired live to the Flask API
- Created the state × material heatmap using pure HTML/CSS colour-intensity mapping
- Implemented dark mode toggle and fully responsive layout for mobile and desktop
- Debugged all file path, API connection, and CORS configuration issues during development
- Set up project folder structure, requirements.txt, and README.md
- Prepared the PowerPoint presentation

7.2 Vinayak Vemula - MS in Data Science (ID: 50124145)

- Identified and collected data from EPA, 8 state agencies, and OECD
- Built the complete data preprocessing pipeline (backend/data_preprocessing.py)
- Performed full exploratory data analysis in Jupyter (recycling_eda.ipynb) - 22 code cells, 8 sections, 12 output charts
- Applied linear regression, Pearson correlation, box plots, histograms, and scatter plots
- Analyzed bottle deposit law policy impact across states
- Wrote all statistical interpretations, key findings, and DATA_SOURCE.md documentation
- Compiled the EDA Report

7.3 Jointly Completed

- Decided which charts and findings to highlight in the dashboard after reviewing EDA results
- Conducted end-to-end testing across all four dashboard tabs and all API endpoints
- Final presentation preparation and rehearsal

8. Limitations and Future Work

8.1 Limitations

- National EPA data ends at 2018 - post-2018 policy changes and COVID-19 impacts are not captured
- Regional data is a single-year snapshot (2019–2020), making year-over-year state comparison impossible
- Global data measures overall MSW rate, not material-specific rates, making direct comparison approximate
- Regional dataset covers only 8 states - sufficient for illustration but not representative of all 50

8.2 Future Work

- Incorporate post-2018 EPA data as it becomes available
- Expand regional coverage to all 50 states for a complete US picture
- Deploy the dashboard to a public URL using Render or Railway (free cloud hosting)
- Add economic analysis: cost-per-ton-recycled by state and material

9. Conclusion

This project successfully met all requirements defined in the project scope: data collection from public sources, preprocessing, exploratory data analysis, regional and material comparison, chart generation, and an interactive HTML dashboard. The final deliverable goes beyond the minimum requirements by including a REST API backend, automated data pipeline, three geographic levels of data, and statistically validated findings.

The most important conclusion is that recycling outcomes are driven by policy and infrastructure, not public awareness alone. The 51-percentage-point gap between Michigan and Texas is explained almost entirely by the presence or absence of a bottle deposit law. The US ranking of 25% globally reflects the fragmented, voluntary nature of US recycling regulation compared to mandatory frameworks used by leading nations. Both team members contributed substantively from their fields of expertise to deliver a complete, working system.

10. References

Data Sources

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